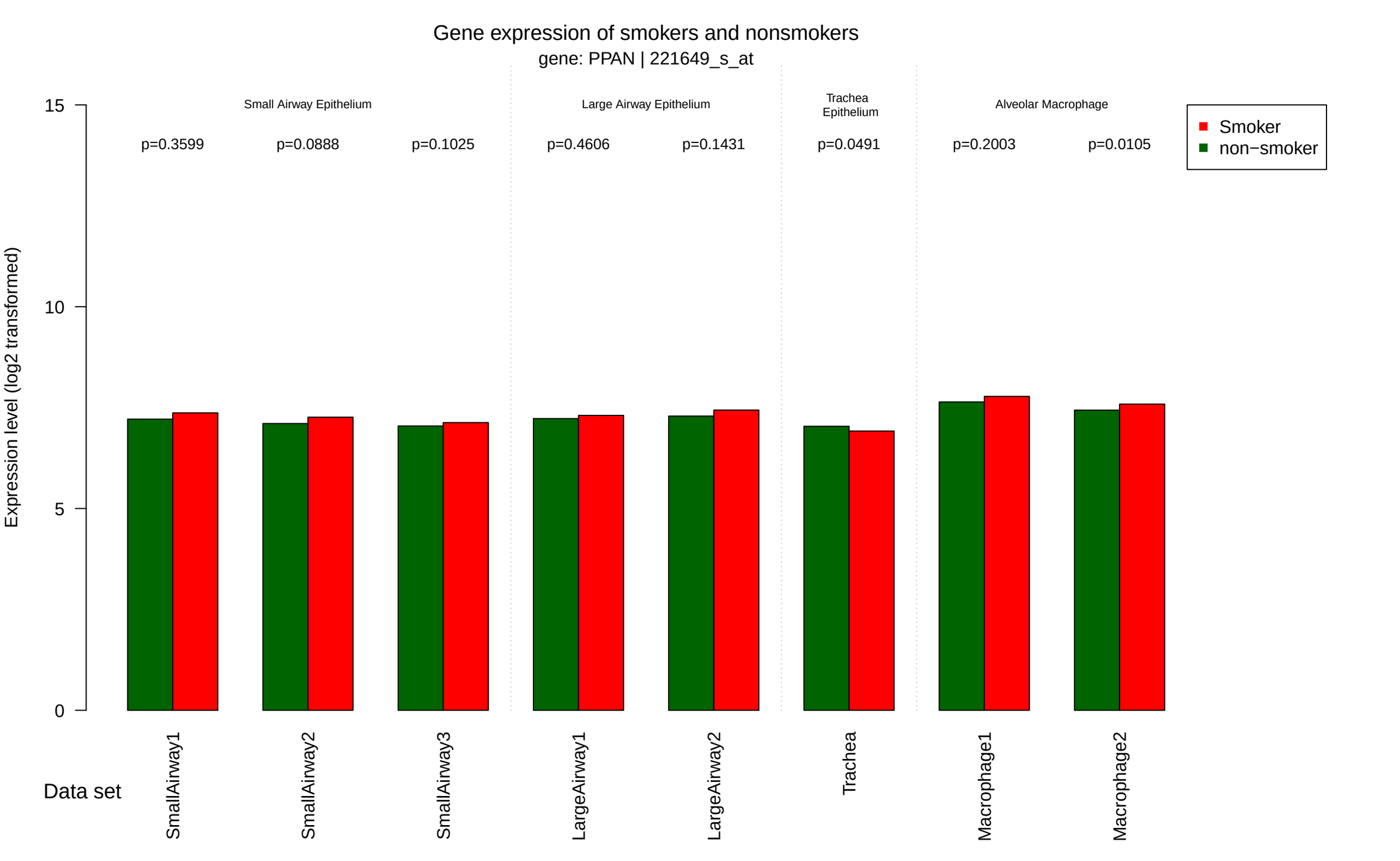
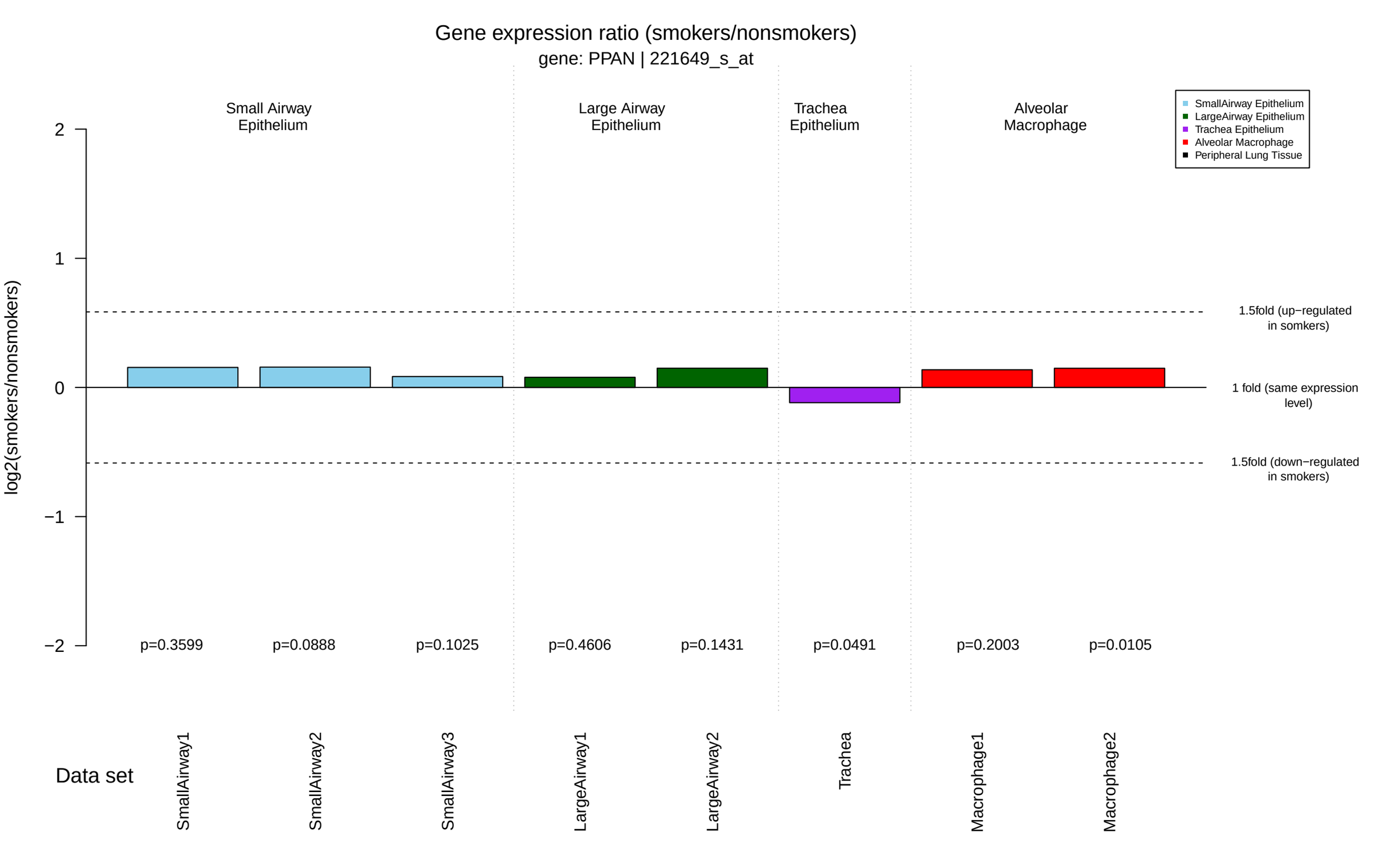
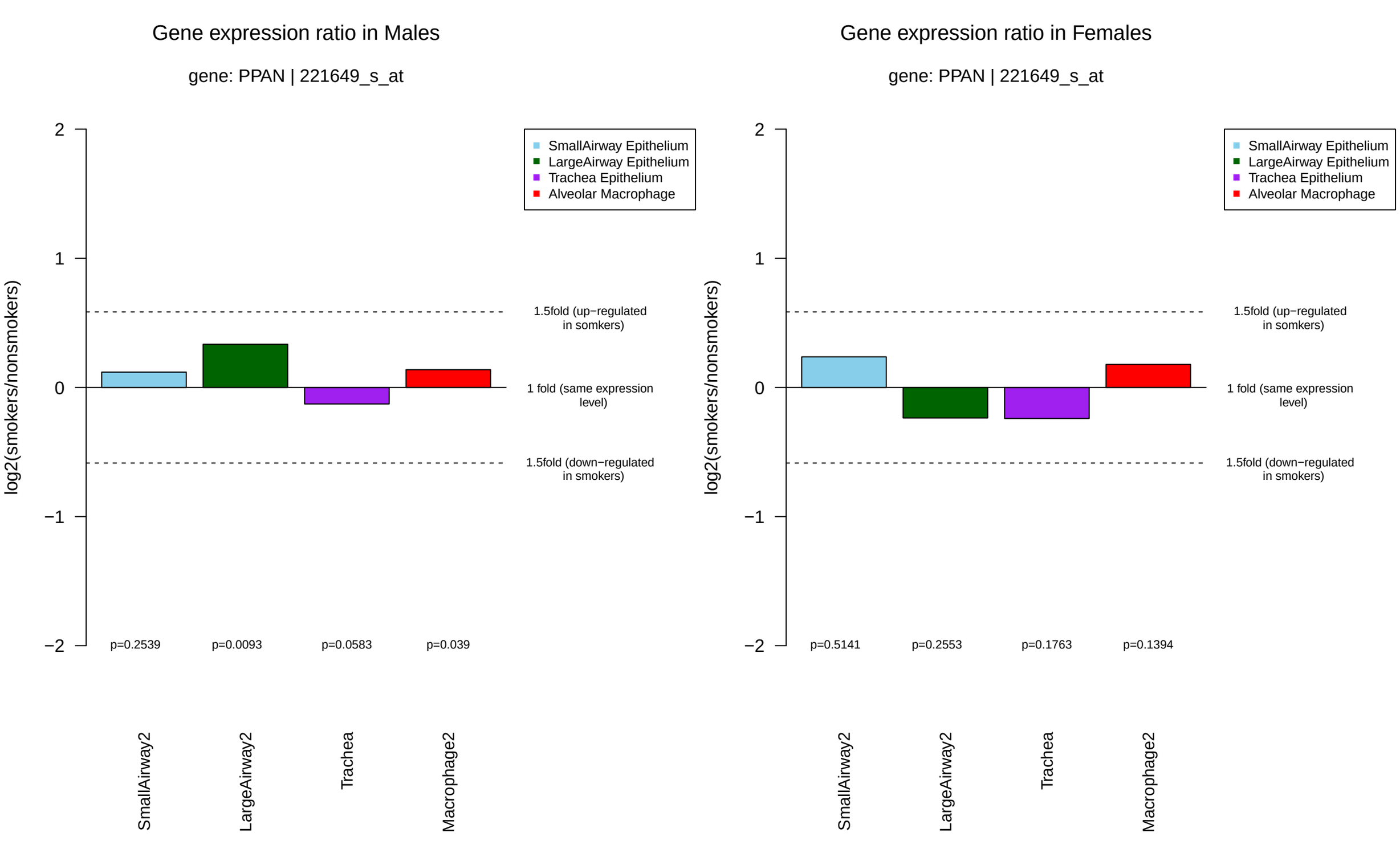
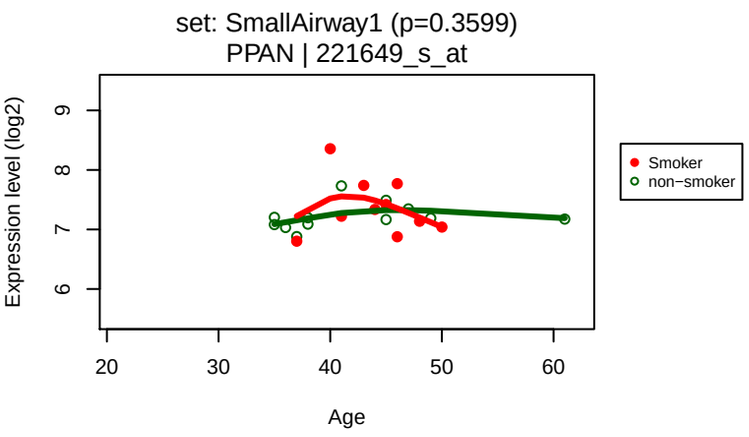
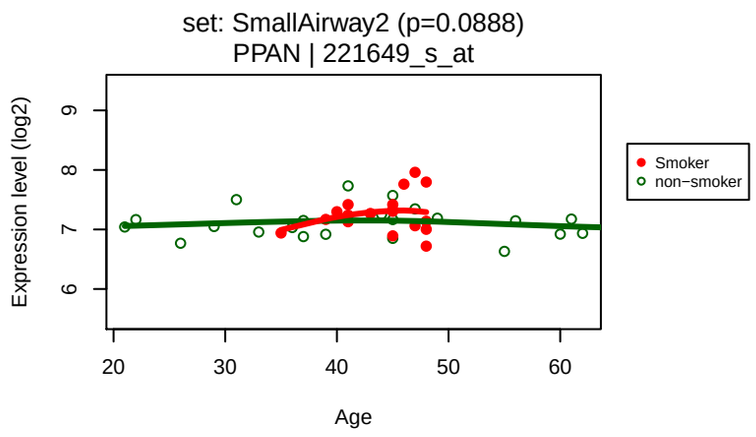
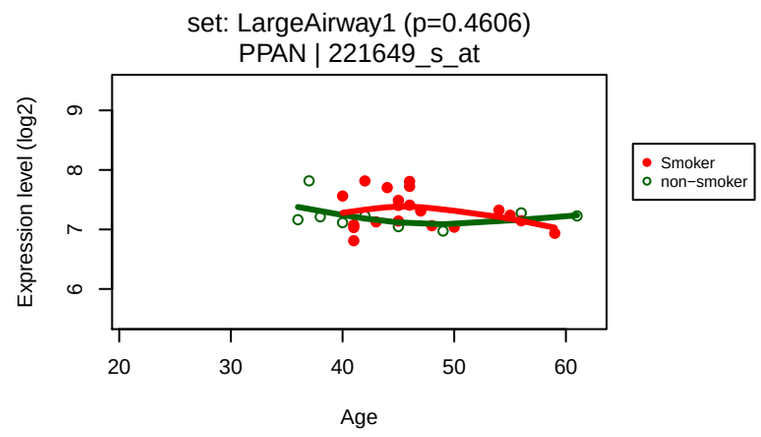

Gene expression of smokers and nonsmokers
gene: PPAN | 221649_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3599
p=0.0888
p=0.1025
p=0.4606
p=0.1431
p=0.0491
p=0.2003
p=0.0105
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

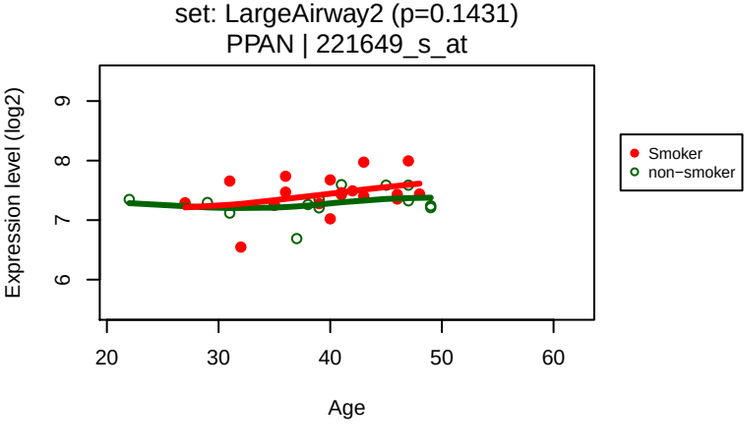

Gene expression ratio (smokers/nonsmokers)
gene: PPAN | 221649_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3599
p=0.0888
p=0.1025
p=0.4606
p=0.1431
p=0.0491
p=0.2003
p=0.0105
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

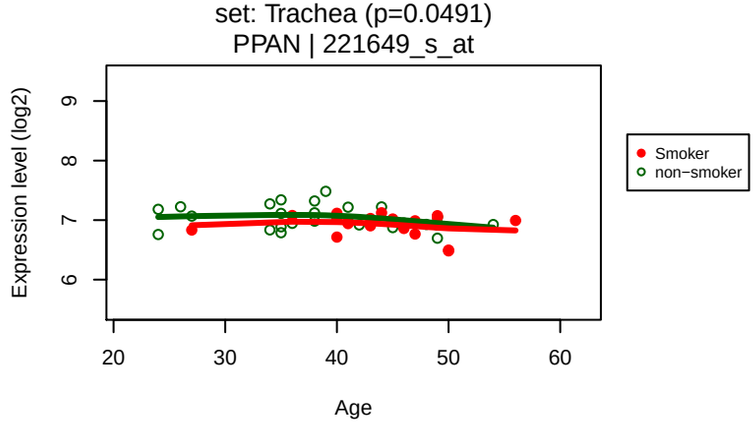

Gene expression ratio in Males
gene: PPAN | 221649_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2539
p=0.0093
p=0.0583
p=0.039
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPAN | 221649_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5141
p=0.2553
p=0.1763
p=0.1394
SmallAirway2
LargeAirway2
Trachea
Macrophage2

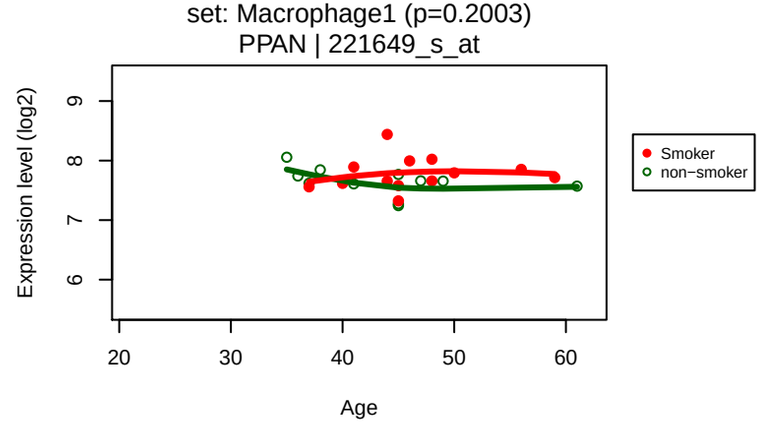

set: SmallAirway1 (p=0.3599)
PPAN | 221649_s_at
Expression level (log2)
Age
Smoker
non−smoker

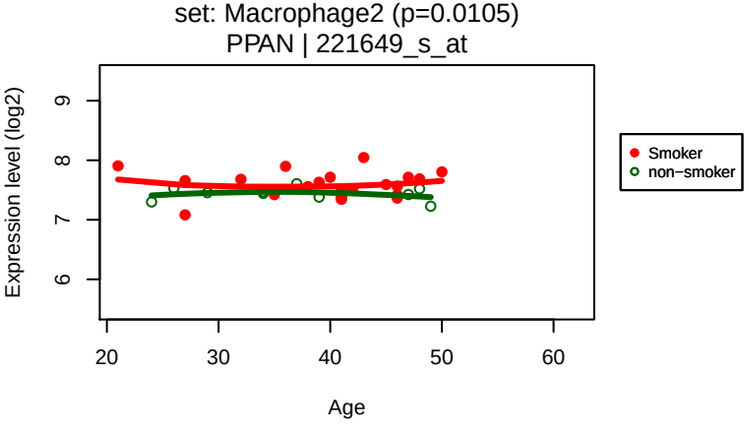

set: SmallAirway2 (p=0.0888)
PPAN | 221649_s_at
Expression level (log2)
Age
Smoker
non−smoker

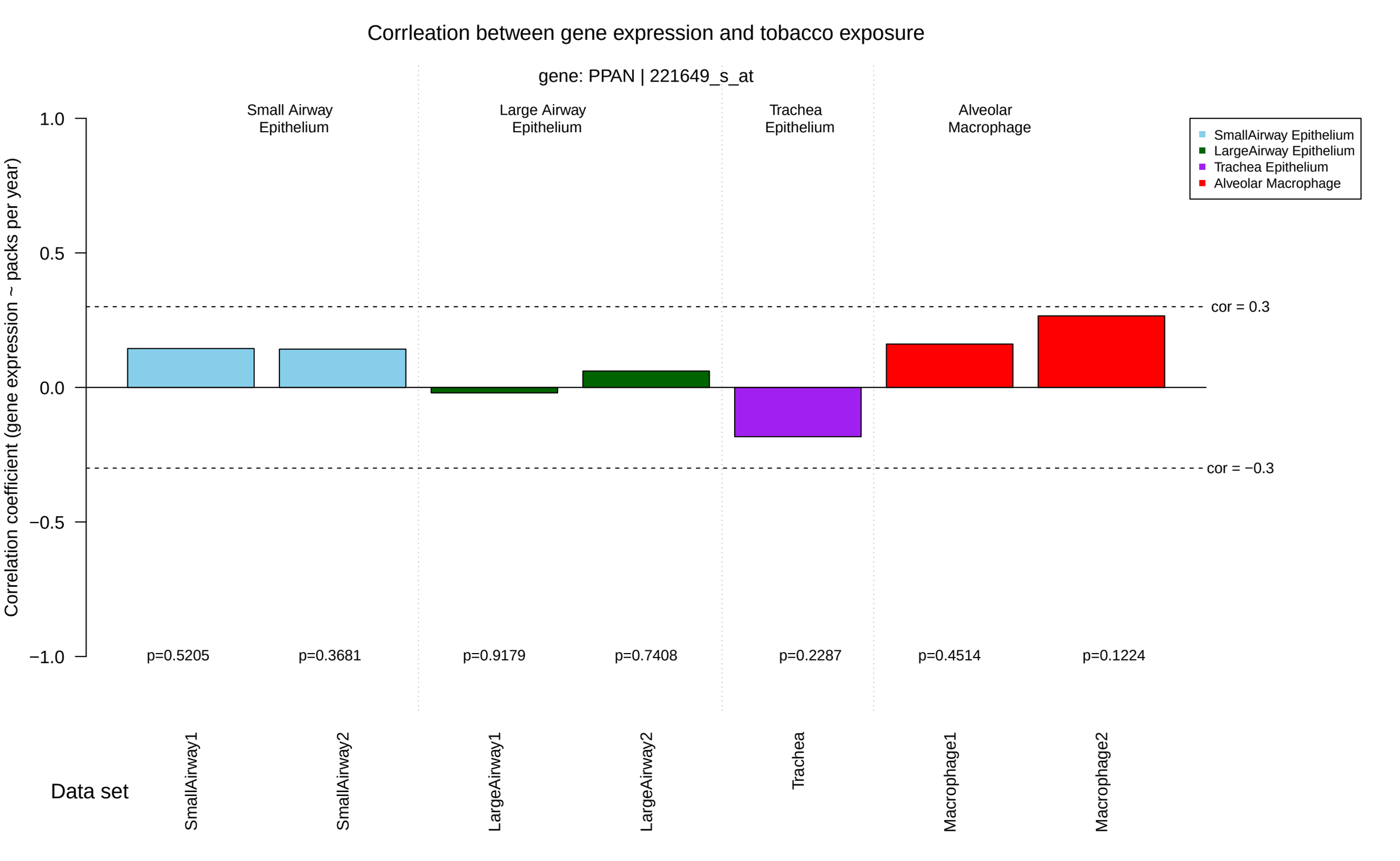

set: LargeAirway1 (p=0.4606)
PPAN | 221649_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1431)
PPAN | 221649_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0491)
PPAN | 221649_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2003)
PPAN | 221649_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0105)
PPAN | 221649_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPAN | 221649_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5205
p=0.3681
p=0.9179
p=0.7408
p=0.2287
p=0.4514
p=0.1224
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2